

IPOPT revisited: Strategies to solve edge cases

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Agenda

Introduction to IPOPT

Techniques applied

Conclusion

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IPOPT: Interior Point OPTimizer

- Paper by Wächter and Biegler in 2005 [1].
- Implementation originally in Fortran (version 1.x and 2.x), now deprecated
- Version 3.x in C++ published in 2006
- Since xxxx part of the COIN-OP foundation [2].
- It is still developed.

IPOPT: Interior Point OPTimizer

One of the most popular open-source optimization packages.

Bindings available in multiple programming languages:

- Python
- Rust
- Fortran?

Inspired other packages and subsequent research

riopt: A Rust re-write from February 2026

Coded in 60/70 hs during the course of 6 weeks heavily relying on Claude Code [3].

According to the benchmarks CUTEst and xxxx, it already beats IPOPT.

Interesting insights into licensing issues and thought process documented in the repo.

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Summary

- Interior Point method
- Inertia correction
- Recovery phase
- Fallback: SQP

Interior Point Methods

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IPOPT's performance can still be improved.

Different methods complement each other: By combining them, edge cases can be solved more efficiently.

Implementation of a state-of-the-art optimization package may be in the reach of modern AI agentic coding tools.

Questions?

Links

Presentation: https://github.com/hlisdero/JMCS_Seminar_Applied_Optimization

Modified argmin: <https://github.com/hlisdero/argmin>

riipop: <https://github.com/jkitchin/riipop>

ipop-rs: <https://github.com/elrnv/ipopt-rs>

Bibliography



A. Wächter and L. T. Biegler, "On the implementation of an interior-point filter line-search algorithm for large-scale nonlinear programming," *Mathematical programming*, vol. 106, no. 1, pp. 25–57, 2006.



C.-O. Foundation, "Coin-op ipopt." <https://github.com/coin-or/Ipopt>.
Accessed: 2026-04-15.



J. Kitchin, "ripopt: A memory-safe interior point optimizer written in rust, inspired by ipopt."
<https://github.com/jkitchin/ripopt>.
Accessed: 2026-04-11.